

SECTION 3 SELECTION AND INSTALLATION OF WIRING SYSTEMS

3.1 GENERAL

3.1.1 Application

This Section specifies the minimum requirements for the selection and installation of wiring systems that shall be achieved to satisfy Part 1 of this Standard.

3.1.2 Selection and installation

Wiring systems shall be selected and installed to perform the following functions or have the following features:

- (a) Protect against physical contact with live parts by durable insulation materials or by placing live parts out of reach.
- (b) Satisfy current-carrying capacity, voltage drop and other minimum size requirements for conductors.
- (c) Provide reliability and electrical continuity of connections, joints and terminations.
- (d) Provide adequate strength of supports, suspensions and fixings.
- (e) Suit intended use, including applications requiring a particular type of wiring system, e.g. fire-resistance, explosion protection, safety services.
- (f) Protect against mechanical damage, environmental and other external influences by enclosure or other means.
- (g) Installed in accordance with the requirements of this Section and the additional requirements as specified in the manufacturer's instructions.

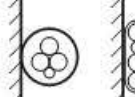
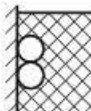
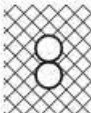
Characteristics of wiring systems that shall be considered include conductor materials, core identification, insulation properties, temperature rise, bending and tension limitations.

3.2 TYPES OF WIRING SYSTEMS

The type of wiring system and method of installation used shall either—

- (a) comply with Table 3.1; or
- (b) have a degree of safety equivalent to that given in Table 3.1.

TABLE 3.1
CABLE TYPES AND THEIR APPLICATION IN WIRING SYSTEMS

Installation method		Description		Typical cable types
		Unenclosed	On a surface (including cable tray or ladder)	Insulated and sheathed Screened or armoured
				
			On a surface partly surrounded by thermal insulation	Mineral insulated, metal sheathed (MIMS) Earthing conductors
			Fully surrounded by thermal insulation	
			Buried direct in the ground, subject to the requirements of Clause 3.11	Insulated and sheathed, screened or armoured, earthing conductors

(continued)

TABLE 3.1 (continued)

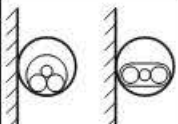
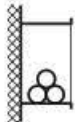
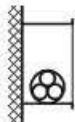
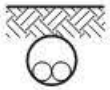
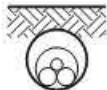
Installation method		Description		Typical cable types
		In an enclosure	On a surface (including cable trunking)	Insulated, unsheathed Insulated and sheathed Screened or armoured MIMS Earthing conductors
			On a surface and partly surrounded by thermal insulation	
			Fully surrounded by thermal insulation	
			Underground, subject to the requirements of Clause 3.11	Insulated, unsheathed Insulated and sheathed Screened or armoured Earthing conductors
				
		Supported on a catenary system		Insulated and sheathed Screened or armoured Earthing conductors
		Supported on insulators		Aerial conductors Earthing conductors

TABLE 3.2
LIMITING TEMPERATURES FOR INSULATED CABLES

Type of cable insulation ⁽¹⁾	Operating temperature of conductor, °C		
	Normal use ⁽²⁾	Maximum permissible ⁽⁷⁾	Minimum ambient ⁽³⁾
Thermoplastic ⁽⁴⁾			
V-75	75	75	0
HFI-75-TP, TPE-75	75	75	-20
V-90	75	90	0
HFI-90-TP, TP-90	75	90	-20
V-90HT	75	105	0
Elastomeric			
R-EP-90	90	90	-40
R-CPE-90, R-HF-90, R-CSP-90	90	90	-20
R-HF-110, R-E-110	110	110	*
R-S-150	150	150	-50
Cross-linked polyethylene			
X-90, X-90UV, X-HF-90	90	90	*
X-HF-110	110	110	*
MIMS ⁽⁵⁾	100	250	(6)
Other types			
PE, LLDPE	70	70	*

* Refer to manufacturer's information.

NOTES:

- 1 The types of cable insulation given in Table 3.2 are included in relevant specifications, i.e. the AS/NZS 5000 series, AS/NZS 3191, AS/NZS 3808 and AS/NZS 60702.1.
- 2 Lower maximum temperatures will apply where materials used in the construction of the cables or in association therewith, such as coverings, sheathings, insulating sleeving on connections and sealing compounds, have maximum operating temperatures lower than the cable proper. However, the allowable operating temperatures for such materials shall not be exceeded.
- 3 If manufacturer's recommendations permit, cables may be installed in locations where temperatures lower than specified may occur.
- 4 The normal operating temperature of thermoplastic cables, including flexible cords installed as installation wiring, is based on a conductor temperature of 75°C. This is because of the risk of thermal deformation of insulation if the cables are clipped, fixed or otherwise installed in a manner that exposes cable to severe mechanical pressure at higher temperatures.